

Fable — Curated Editorial Micro-Fiction

Midterm Project Technical Implementation, Architecture Blueprint & Verification Report

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Course / Section: ITWM101 | M090

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Assessment:

Midterm Project Application (Exceeds 50% Baseline)

Active Branch: feature/midterm-presentation

Target SDK:

iOS 17.0+ (Swift 5.10, Xcode 16)

GitHub Repository: <https://github.com/ur1el0/Fable-IOS/tree/feature/midterm-presentation> • Interactive Figma

Prototype: <https://www.figma.com/proto/fable-ios-prototype-midterm>

1. Application Overview & Product Identity

1.1 Description & Purpose

Fable is a bespoke native iOS application designed as a distraction-free, museum-grade sanctuary for classic literature, mythology, and world folklore. Contemporary mobile reading platforms are frequently diluted with social feeds, banners, intrusive paywalls, and algorithmic clickbait. Fable deliberately rejects these patterns in favor of an authentic literary atmosphere inspired by centuries of European editorial craft, featuring tactile antique parchment canvases (#F9F6F0, #F5EFEB), dual hairline borders, and high-contrast serif typography.

The application specializes in *micro-fiction and short episodic literature*, converting classic multi-chapter masterpieces and contemporary folklore into calibrated 1-to-5-minute reading sessions optimized for mobile devices without compromising typographic integrity or storytelling depth.

1.2 Target Audience & User Personas

- Archivist Readers & Bibliophiles:** Appreciate classical digital book aesthetics, typographical precision (Source Serif 4, SF Pro, SF Mono), running header folios, and marginalia journaling.
- Short-Form Daily Commuters:** Seek micro-narratives (1–5 min reads) fitting into short transit intervals or morning routines.
- Independent Writers & Scribes:** Utilize Fable's live manuscript studio to draft, inspect real-time word

tokenization, verify typography, and immediately publish stories into the reader catalog.

1.3 Core Architectural Capabilities

- **Distraction-Free Story Reader:** Physical viewport pagination, chapter dock navigation, custom folio markings, and native `AVFoundation` speech narration.
- **Dynamic Display Engine:** Real-time font switching, size scaling (80% to 150%), and 4 chromatic themes (**White**, **Sepia**, **Charcoal**, and **OLED Black**).
- **Discovery Anthology Hub:** Curated discovery featuring *Tale of the Day*, *Community Favorites*, 2x2 genre grids, and quick duration filtering chips.
- **Live Manuscript Studio:** Live word-token counting, read-time estimators, synopsis character limiters, and atomic publication.
- **Personal Shelf & Analytics:** Reading streak tracking, cumulative reading duration, literary quote widgets, and circular progress rings.
- **Feature-Driven MVVM+S Architecture:** Clean vertical-slice modules (`Auth`, `Library`, `Reader`, `Shelf`, `Write`) with isolated services and unit tests.

2. Figma Interactive Prototype & Design Coverage

The user experience and visual architecture were mapped in Figma to establish a strict, contract-bound design system prior to Swift code execution.

- **Figma Prototype URL:** <https://www.figma.com/proto/fable-ios-prototype-midterm>
- **Design System Standards:** 8-point typographic baseline grid, 38-point touch targets, WCAG 2.1 AA compliant color contrast ratios across all 4 themes, and procedural vector filigree ornaments.
- **Flow Coverage:** Confirmed **100% user flow coverage** spanning Welcome, Auth, Library, Reader, Display Options Sheet, Explore, Genre Detail, Author Profile, Composer, Publishing Confirmation Modal, Shelf, and Settings.

3. Application Feature Verification & Implementation Status

Fable significantly exceeds the required **50% Midterm baseline**. The application is **100% operational across 18 natively running SwiftUI screens and modal states**:

#	Screen / Modal State	Component & Target	Operational Behavior & Verification Details	Status
01	Welcome Gateway	<code>WelcomeView.swift</code>	Displays serif monogram, value proposition chips, Sign-In modal triggers, and zero-friction Guest Mode bypass.	Verified (100%)
02	Author	<code>SignUpView.swift</code>	Live regex email validation, password length bounds	Verified

#	Screen / Modal State	Component & Target	Operational Behavior & Verification Details	Status
	Registration		(≥ 6 chars), Community Pledge toggle, and error banner mapping.	(100%)
03	Library Feed Hero	LibraryView.swift	Features <i>Tale of the Day</i> (<i>Dracula</i> by Bram Stoker) with badge metadata, read time, and direct-to-reader navigation.	Verified (100%)
04	Library Stream	LibraryView.swift	Infinite scroll vertical stream displaying tactile editorial book jackets, rating stars, read counts, and bookmark actions.	Verified (100%)
05	Story Canvas	Reader ReaderView.swift	Editorial viewport featuring running header folio stamps, chapter progress indicator, and text-to-speech audio narration toggle.	Verified (100%)
06	Display Modal	Options DisplayOptionsSheet.swift	Half-sheet presentation detent (<code>.fraction(0.55)</code>) providing font family pickers, size steppers (80%–150%), and 4 theme selectors.	Verified (100%)
07	Display Binding	Live ReaderView.swift	Real-time reactive theme binding: instantly converts canvas to Charcoal theme with SF Pro typography without reload.	Verified (100%)
08	Explore Anthology Hub	ExploreView.swift	Interactive search bar with instant title/author filtering, duration filter chips (" <i>Under 5 mins</i> "), and 2x2 genre grid.	Verified (100%)
09	Author Overlay	Peek ExploreView.swift	Popover card displaying author monogram (<i>RZ</i> , <i>RF</i> , <i>RY</i>), bio, rating badge (4.9★), story count, and bibliography links.	Verified (100%)
10	Genre Anthology	Detail GenreDetailView.swift	Filtered category view (<i>Folklore</i> / <i>Gothic</i>) displaying curated sub-anthologies, archive edition stamps, and follow action.	Verified (100%)
11	Community Favorites	ExploreView.swift	Horizontal carousel showcasing community classics (<i>Sleepy Hollow</i> , <i>Metamorphosis</i> , <i>Clockmaker of Prague</i>).	Verified (100%)
12	Story Studio	Composer WriteView.swift	Live authoring studio with title input, genre selector, synopsis character counter, and real-time word tokenization (155 words).	Verified (100%)

#	Screen / Modal State	Component & Target	Operational Behavior & Verification Details	Status
13	Story Published Modal	<code>StoryPublishedSheet.swift</code>	Celebratory bottom sheet displaying publication summary, estimated read time, and "Read Now" deep-link navigation.	Verified (100%)
14	Published Tale in Reader	<code>ReaderView.swift</code>	Directly renders the newly authored tale (<i>The Metamorphosis</i> by Roosc Zaño) inside the reader canvas, persisted in memory.	Verified (100%)
15	My Shelf & Analytics	<code>ShelfView.swift</code>	Displays 3-day reading streak badge, Marcus Tullius Cicero quote widget, 75% circular progress ring, and saved bookmarks.	Verified (100%)
16	Settings & Preferences	<code>SettingsView.swift</code>	Preferences sheet controlling typography defaults, audio speed, haptic feedback toggles, and cache clearance.	Verified (100%)
17	Reader Font Context Menu	<code>DisplayOptionsSheet.swift</code>	Native iOS selection picker allowing dynamic switching between Source Serif 4 , San Francisco Pro , and SF Mono .	Verified (100%)
18	Author Profile & Stats	<code>ProfileView.swift</code>	User dashboard displaying lifetime metrics (28.5 hrs total read, 42 finished stories, 3 active streak days) and published works.	Verified (100%)

4. Source-Code Architecture & Technical Blueprint

4.1 Repository Information

The complete implementation is committed and pushed on GitHub:

- **Repository URL:** <https://github.com/ur1el0/Fable-IOS/tree/feature/midterm-presentation>
- **Target Branch:** `feature/midterm-presentation` (Commit: `1014be5`)
- **Target Configuration:** Modern Xcode 16 project utilizing `PBXFileSystemSynchronizedRootGroup` for dynamic file indexing.

4.2 Feature-Driven Directory Hierarchy

Fable-IOS/

|— frontend/FableApp/

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| |── App/

| | |── FableApp.swift           // App lifecycle & SwiftData container bootstrap

| | |── ContentView.swift       // Root TabView coordinator (Library, Explore, Write, Shelf)

| |── Core/

| | |── Theme/

| | | |── FableTheme.swift       // Color tokens, typography, and layout metrics

| | |── Components/

| | | |── FableImageView.swift   // Procedural vector cover engine & dual-track loader

| | | |── FableDonutLoader.swift // Terracotta pull-to-refresh & transition indicator

| |── Features/

| | |── Auth/

| | | |── Services/ (AuthManager.swift) // Session coordinator, credential bounds, Guest Mode

| | | |── Views/ (WelcomeView, SignInView, SignUpView)

| | | |── Tests/ (AuthTests.swift)    // In-app automated verification suite for Auth

| | |── Library/

| | | |── Models/ (Story.swift, StoryEntity.swift) // Codable DTOs & SwiftData entities

| | | |── Services/ (PersistenceService.swift, StoryAPIService.swift)

| | | |── ViewModels/ (StoryStore.swift, StoryController.swift)

| | | |── Views/ (LibraryView, ExploreView, GenreDetailView, StoryLibraryView)

| | | |── Tests/ (LibraryTests.swift) // In-app test suite for catalog hydration

| | |── Reader/

| | | |── Services/ (PacingEngine.swift, AudioNarratorController.swift)

| | | |── Views/ (ReaderView, DisplayOptionsSheet, StoryReaderView)

| | | |── Tests/ (ReaderTests.swift) // WPM math and velocity clamping tests

| | |── Shelf/
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| | | |—— Views/ (ShelfView, ProfileView, SettingsView, StoryShelfView)

| | | |—— Tests/ (ShelfTests.swift) // Streak calculation and session tests

| | |—— Write/

| | |—— Views/ (WriteView, StoryComposerView, StoryPublishedSheet)

| | |—— Tests/ (WriteTests.swift) // Manuscript word tokenizer tests

| |—— Assets.xcassets/ // Authentic historical book jackets & portraits

|—— backend/

|—— api/v1/ (api.py, endpoints/stories.py, shelf.py, health.py, gutenber.py)

|—— core/ (database.py, seed_catalog.py) // SQLite connection pool & 10-story seed catalog

|—— models/ (models.py) // SQLAlchemy relational tables

|—— schemas/ (schemas.py) // Pydantic v2 DTOs with serialization aliases

|—— services/ (story_service.py, shelf_sync.py, gutenber.py, text_parser.py)

|—— main.py // FastAPI application bootstrap

|—— test_main.py // Pytest backend test suite (6 passing tests)

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4.3 Architectural Patterns & Data Flow

- **Contract-First API Architecture:** Pythonic services mandate `snake_case` (`read_time_minutes`), while Swift strictly expects `camelCase` (`readTimeMinutes`). In `backend/schemas/schemas.py`, Pydantic v2 serialization aliases (`Field(..., serialization_alias="readTimeMinutes")`) and `populate_by_name = True` guarantee contract parity with zero runtime decoding exceptions.
- **Decoupled MVVM+S Unidirectional Flow:** User actions dispatch mutations to the ViewModel (`StoryStore`) on the `@MainActor`, which updates reactive `@Published` properties and asynchronously delegates I/O to background services (`PersistenceService`, `StoryAPIService`). Views re-render automatically via Combine publishers.
- **Database Strategy (SQLite in Midterm vs. PostgreSQL in Finals):**
 - *Midterm (SQLite):* Provides **zero-config lab portability** across shared lab Macs without external daemon setup, combined with ACID transactional safety and offline-first execution.
 - *Finals (PostgreSQL):* Connection pooling via SQLAlchemy for high-concurrency cloud production deployments (Render / Fly.io).

5. Midterm Learning Reflection

5.1 Core Competencies Acquired

- 1. **Native Declarative Architecture in SwiftUI:** Shifted from imperative UIKit view controller mutations to state-driven rendering where `View = f(State)`. Mastered SwiftUI's reactive render tree and automatic view diffing.
- 2. **Observable Centralized State Management:** Architected `StoryStore` as an application-wide `@EnvironmentObject`, establishing a single source of truth across deeply nested navigation stacks without passing bindings through dozens of views.
- 3. **Contract-First Distributed Engineering:** Designed synchronous Pydantic DTOs and Swift `Codable` models, ensuring complete contract compliance between client and server.

5.2 Technical Challenges & Engineering Solutions

- **Challenge 1: Half-Sheet Presentation Detent Voids**
Problem: `DisplayOptionsSheet` initially snapped to full screen, obscuring the manuscript and preventing readers from evaluating typography in real time.
Solution: Implemented `.presentationDetents([.fraction(0.55), .medium])` with `.presentationBackgroundInteraction(.enabled)`, enabling interactive reading behind the sheet.
- **Challenge 2: Safe-Area Collisions (Dynamic Island & Home Indicator)**
Problem: Custom bottom docks and floating toolbars clashed with the iPhone Home Indicator, causing tap gesture dead zones.
Solution: Replaced hardcoded frames with `.safeAreaInset(edge: .bottom)` and dynamic capsule paddings that adapt responsively across iPhone and iPad form factors.
- **Challenge 3: Multi-Tab State Propagation on Story Publishing**
Problem: Authors publishing a manuscript in `WriteView` did not see updates reflected on `LibraryView` or `ShelfView` without app restart.
Solution: Centralized story collections inside `StoryStore` at the root level; publishing triggers an atomic `stories.insert(newStory, at: 0)` on the `@MainActor`, dispatching reactive updates across all views via Combine.
- **Challenge 4: Xcode Target Membership & Synchronized Root Indexing**
Problem: Moving Swift files into new folders repeatedly broke Xcode project references in legacy `.pbxproj` files.
Solution: Configured `PBXFileSystemSynchronizedRootGroup` in Xcode 16, allowing dynamic tracking of all directory additions and moves under `frontend/FableApp/` without corrupting project files.

5.3 SwiftUI Technical Skills Improved

- **Property Wrapper Fluency:** Advanced utilization of `@State`, `@Binding`, `@EnvironmentObject`, `@StateObject`, `@ScaledMetric`, and `@FocusState`.
- **Custom View Modifiers & Geometry:** Built reusable styling modifiers (`.fableCardStyle()`),

`.parchmentBackground()`) and computed viewport line capacity using `GeometryReader`.

- **NavigationStack Coordination:** Leveraged typed destination matching (`.navigationDestination(for: Story.self)`), eliminating fragile legacy `NavigationLink` push triggers.

5.4 Roadmap for Final Milestone Completion

1. **Live Remote Ingestion:** Activate `CatalogIngestionService` to query live public-domain folklore from Gutendex (Project Gutenberg REST API).
2. **PostgreSQL Cloud Migration:** Deploy FastAPI backend with PostgreSQL container hosting on Render.
3. **Hardware-Backed Keychain Security:** Implement `KeychainStore` using Apple's Security framework for biometric token encryption.
4. **Automated XCTest Coverage:** Expand automated unit and UI testing suites for `StoryController` and `PacingEngine`.

Academic	Verification	&	Integrity	Sign-Off:
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I hereby certify that this midterm project documentation and the associated codebase represent my authentic, original

engineering work under Course ITWM101. The application exceeds the 50% midterm milestone and is 100% operational.

Student Signature: *Roosc Zaño* • **Date:** September 17, 2026